PM Graded Quiz M1

Question 1

Would the mean-variance optimization problem be the same for two investors who have different utility functions (say one investor has a quadratic utility function and another a log utility function like $U(w) = \ln(1 + w)$)?

- Yes, if the assets that form the portfolios have a multivariate normal distribution.
- No, they have two very different utility functions and solutions can never be the same.
- Yes, because the log-utility function is for a risk taker, and the quadratic utility function describes a risk-averse investor.
- Yes, the utility function is only marginally relevant to define the problem.

Question 2

Suppose that a portfolio consists of three assets and that the covariance matrix, Σ, for these assets is:

$$\begin{pmatrix} 1.0 & -0.5 & 0.6 \\ -0.5 & 2.0 & 0.8 \\ 0.6 & 0.8 & 3.0 \end{pmatrix}$$

What is the variance of the naive portfolio (i.e., a portfolio that assigns the same weight to each asset)?

- 7.80
- 3.69
- 0.867
- 2.00

Question 3

Given that a $10,000,000 portfolio achieved the following returns over a period of 20 months :

-2.1%	-2.45%	-2.3%	-2%	-2%
-1.8%	-1.8%	-1%	0.5%	0.6%
0.9%	1%	1.4%	2%	2.1%
2.3%	2.7%	2.9%	3%	3.5%

What is the 90% one-month expected shortfall using a nonparametric approach (use nearest-rank method to determine the percentile)?

- $237,500
- $230,000

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- $210,000
- (•) $245,000

Question 4

Assume that a portfolio has a correlation of 0.5 with the market portfolio. In addition, we are told that the volatility of the portfolio is 20%. What is the variance of the unsystematic part of the risk? What would the volatility of the unsystematic part of the risk be if the correlation were -0.75?

- 3% and 13.23%, respectively
- 15.00% and 8.75%, respectively
- 15.00% and 13.23%, respectively
- (•) 10.00% and 5.00%, respectively

Question 5

Consider this game: A gambler flips a coin, and if it is heads (H), she receives $3. If it turns out tails (T), she flips again and loses $1. In the second toss, If the coin comes up H, now she wins $6, and if it turns out T, she loses $2.

The gambler's utility function is U(w) = (w-3)^2 and her initial wealth is $3.

What is her expected utility?

- 10.75
- 20
- 13
- (•) 5

Question 6

The efficient frontier was derived under a single constraint, i.e., the sum of weights is 1. What would happen to the efficient frontier if we imposed additional constraints on the problem, say we do not allow any stock to be more than 20% of the overall portfolio?

- The efficient frontier would not move at all since it does not respond to the number of constraints.
- The efficient frontier would move up since by limiting the exposure to single assets, we would reduce the risk.
- The efficient frontier would move to the right and be lower than the current one.
- (•) The efficient frontier moves to the left as we can now achieve the same return with less risk.

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Question 7

Which of the following best describes a linear portfolio?

- European call and put options are linear functions of their prices and should be included in a linear portfolio.
- Government bonds are a linear function of the underlying interest rates and should be in a linear portfolio.
- Exotic options always have small gammas, so they can be considered linear assets with respect to their underlying prices.
- ◉ Asset values are a linear function of their prices.

Question 8

The following data has been collected to appraise the performance of four asset management firms:

	Fund A	Fund B	Fund C	Fund D	Market Index
Return	6.45%	9.10%	9.44%	5.82%	7.50%
Beta	0.8	1.05	1.36	1.80	1.00
Std. dev.	2.75%	4.55%	3.72%	2.65%	2.82%

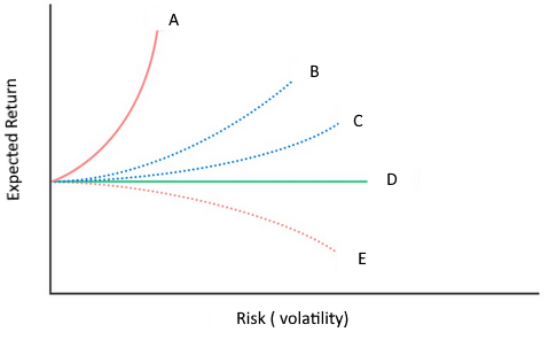
The risk-free rate during the reference period was 3%.

- What is the rank of the four funds based on the M2 measure (from best to worst)?
- Would the ranking be different if we had used the Sharpe ratio?

- Ranking: C, B, A, D; No.
- Ranking C, B, A, D; Yes.
- Ranking C, B, D, A; No.
- ◉ Ranking: C, B, A, D; Yes, the ranking can be different if we use different measures of appraisal.

Question 9

Look at the following plot showing five different investors and their level of indifference between risk and expected return:



A) B

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B) E

C) D

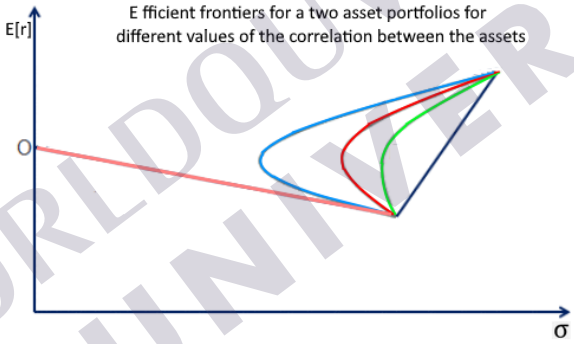
D) C

E) A

Moderately Risk-Averse	A
Risk Seeker	B
Very Risk-Averse	C
Risk-Neutral	D

Question 10

Assume that you have two assets in your portfolio. Each of the lines in the figure represent an efficient frontier depending on the value of the correlation between the two assets. Which one is the efficient frontier for $\rho = 1$?



- (●) The blue line
- ( ) The black line
- ( ) The orange line
- ( ) The green line

Question 11

If the monthly volatility is 1.5%, what is the annualized volatility?

- ( ) 18.00%
- ( ) 2.70%
- (●) 0.27%
- ( ) 5.196%

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Question 12

A wealthy client is considering investing a substantial amount of money. She has narrowed her choices down to two portfolio managers, A and B. The minimum return she finds acceptable in any year is 5%.

	Manager A	Manager B
average return	18%	22%
volatility	15%	15%
downside volat.	6%	8.5%

The risk-free rate of return is 2%.

- What manager should she choose if she adopted the Sortino ratio as a measure to decide?
- Do you think that the Sharpe ratio would be a better choice?

- Manager A; No, because the Sharpe ratio would punish portfolio managers for both "good" and "bad" volatility.
- (•) Manager B; Yes, the Sharpe ratio is a superior measure.
- Manager A; Yes, the Sharpe ratio is a superior measure.
- Manager B; No, the Sharpe ratio would punish portfolio managers for both "good" and "bad" volatility.

Question 13

Suppose that Antonio is an investor. His Arrow-Pratt measure of absolute risk is the following:

$$A(w) = \frac{2\alpha}{1 - 2\alpha w}$$

where $\alpha > 1/2$. What can we say about Antonio's absolute aversion to risk when his wealth, w, is such that $w > \frac{1}{2\alpha}$?

- It increases with wealth.
- (•) Antonio's risk aversion increases with wealth and goes to infinity as his wealth gets larger and larger.
- It increases with wealth. However, after he reaches a certain level of wealth, Antonio becomes less risk averse.
- Since w appears in the denominator, his risk aversion is increasing.

Question 14

Why is the two-fund separation theorem useful?

- Because by using it, we can avoid quadratic programming completely

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QUESTIONS

1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

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- Because it allows us to derive the efficient frontier once we have two portfolios with different expected returns that lie on the efficient frontier.
- (•) Because it allows us to derive the efficient frontier once we have two portfolios that lie on the efficient frontier
- Because it provides a good enough approximation compared to using quadratic programming to derive the efficient frontier, and it is therefore computationally convenient.

Question 15

Which of the following statements is **false?**

- Exponential weighted moving average (EWMA) is a methodology to deal with non-constant variance when calculating VaR.
- (•) Garch models for variance are useful when computing VaR for assets that exhibit volatility clustering.
- Parametric VaR never yields results that are very different from non-parametric VaR
- The non-parametric/historical VaR makes no assumption about the underlying distribution of returns.

Question 16

A portfolio manager has reported a Sharpe ratio of 1.3. The skewness of the return of his portfolio is 0.5, and the kurtosis is 4. What is the adjusted Sharpe ratio for this portfolio manager?

- 1.349
- (•) 1.008
- 1.260
- 1.392

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